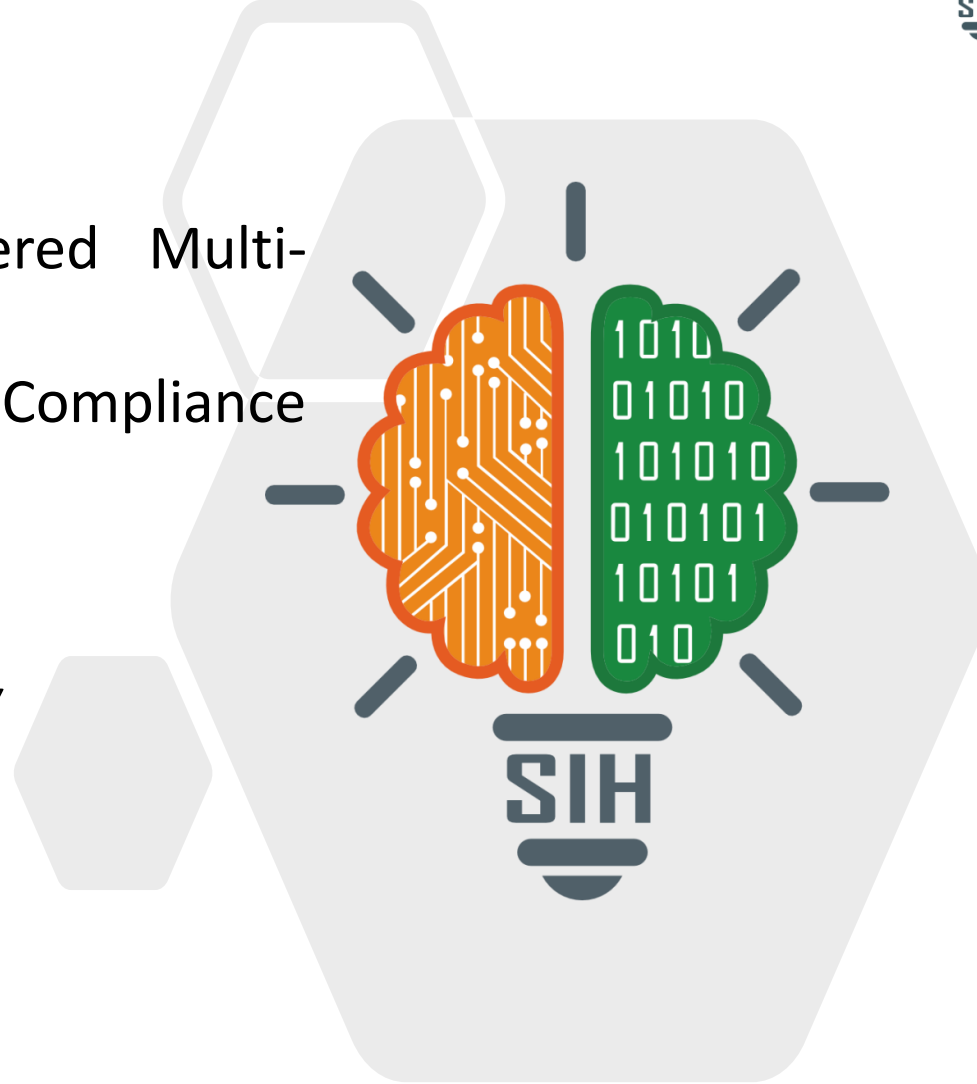


SMART INDIA HACKATHON 2026



- **Problem Statement ID** – SIH26155
- **Problem Statement Title-** AI-Powered Multi-Vendor Network Configuration, Security Compliance and Risk Audit Platform
- **Theme-** BLOCKCHAIN & CYBERSECURITY
- **PS Category-** Software
- **Team ID-**
- **Team Name** - NEXTGEN



The Challenge

Sandir-Security Standards, Different Configurations.

Security compliance rules are standardized (e.g., CIS / NIST / STIG), but network device configurations are not. Different vendors use different CLI syntax and structures, making manual compliance analysis difficult error-prone and time-consuming



Different vendors, different CLI syntax and configuration structures



Manual analysis is slow, expertise-dependent and inconsistent



Unmapped or unfamiliar configuration lines create uncertainty



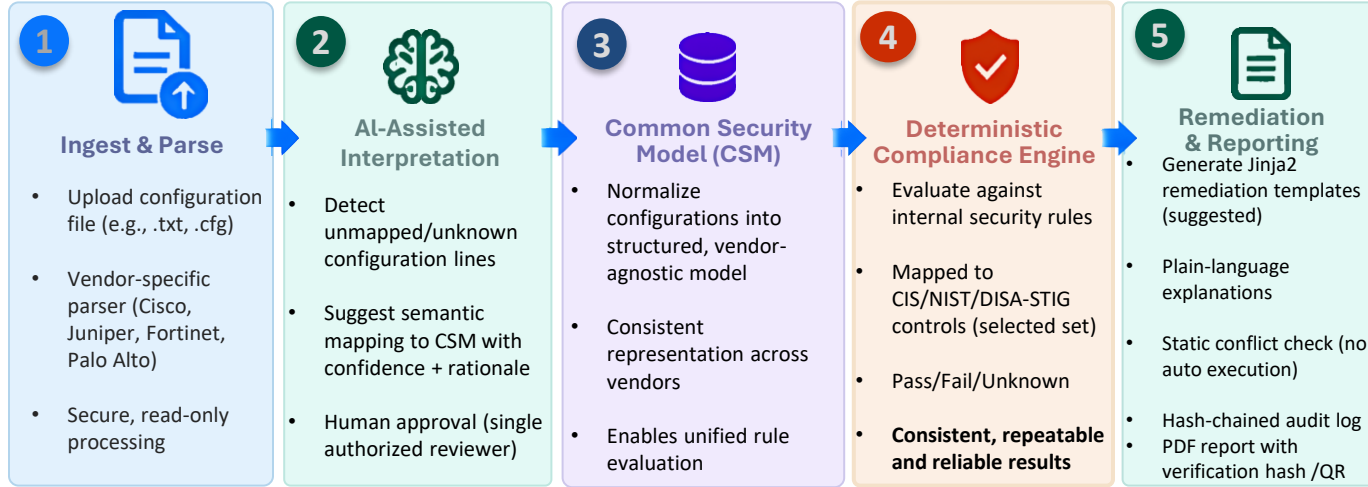
High risk of misconfigurations and security gaps in critical infrastructure

Need a unified, automated and reliable approach for consistent security compliance across diverse network vendors.

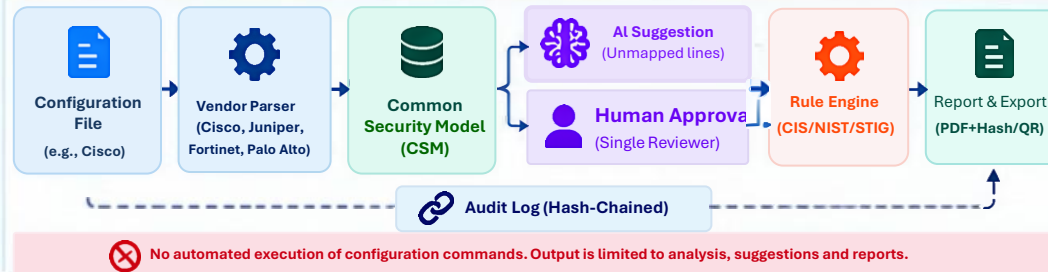


OUR SOLUTION

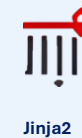
Unified, AI-Assisted and Deterministic Compliance



Architecture Overview (MvP Scope-Single File, Single Path)



Tech Stack (MVP)



Deterministic & Explainable

Rule-based engine evaluates against CIS /NIST /STIG with clear rationale and consistent results.



Single Authorized Reviewer (MVP)

Simplified and secure approval process. All suggestions are validated before use.



AI Assistance, Not AI Control

AI interprets unmapped configurations and suggests mappings. Human approval is required. No AI output directly decides compliance or executes commands.



Vendor-Neutral Normalization

Converts diverse vendor configurations into a common security model (CSM).



What Makes It Different?

A Practical, Secure and Vendor-Neutral Approach



MVP Validation (Target /Measured on Reference Setup)



Deterministic Evaluation

<5 seconds
for ~2,000 lines
(4-core CPU/8GB RAM)



Compliance States

Pass/Fail/Unknown
demonstrated



AI Mapping + Approval

End-to-end workflow
working



Reproducible Results

Same input → same output
(5+ runs)



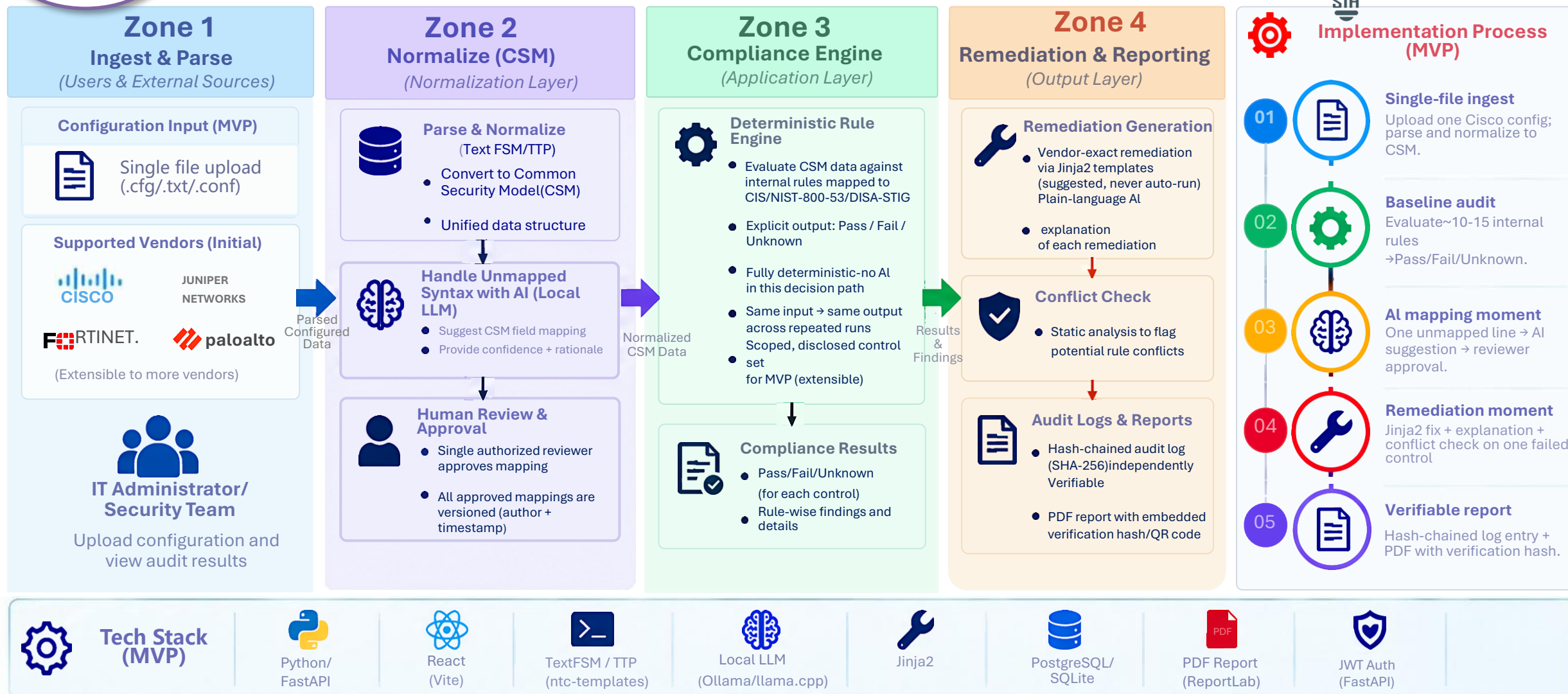
Future Scope

- More vendors and controls
- Web-based multi-user platform**
- Integration with enterprise workflows

TECHNICAL APPROACH



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2026**

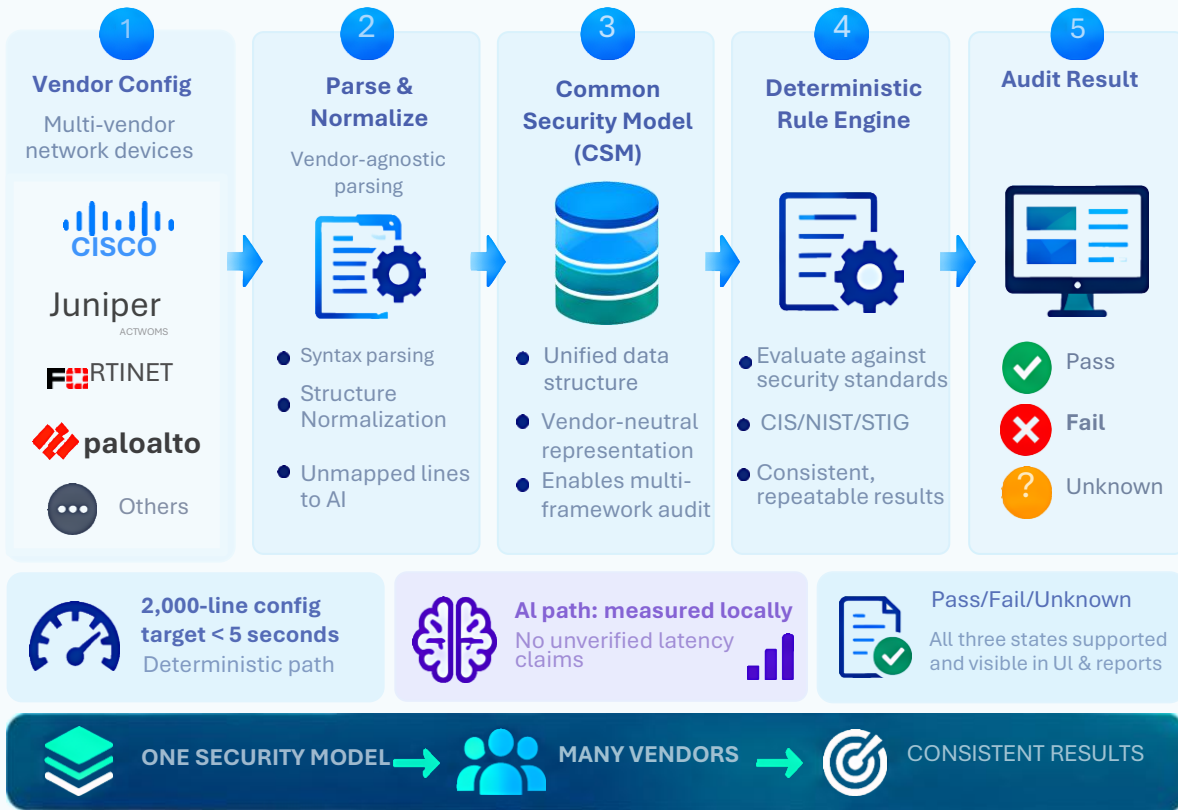


No configuration command is executed automatically anywhere in this flow — output is limited to analysis, suggestions, and reports.



TECHNICAL FEASIBILITY

Proven AI + Deterministic Architecture



OPERATIONAL FEASIBILITY

Practical • Secure • Implementable



ECONOMIC VIABILITY

Cost-Effective • Scalable • Sustainable

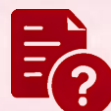


KEY RISKS



MITIGATIONS

Anticipate • Prevent • Build with Confidence



Unknown/Vendor Syntax

Mitigation: AI-assisted interpretation with human-approved mapping.



Bad Remediation/ Conflicts

Mitigation: Static analysis + simulation to detect conflicts before recommendation.



False Compliance Mitigation:

Deterministic rules + auditable evidence (logs, hashes, reports).



IMPACTS

Why This Problem Matters

Current State

Manual | Fragmented | Risk-Prone



Increased Security Risk

Misconfigurations and policy deviations can lead to security breaches in critical infrastructure

74%+

breaches involve the human element — including error and misconfiguration (Verizon DBIR, 2023)



Time-Consuming Audits

Manual analysis of large configuration files is slow and requires expert knowledge

Days-Weeks

per device for manual audit and review



Inconsistent Compliance

Different vendors, OS, syntax and structures make uniform compliance difficult.

Multiple interpretations

for the same control across vendors



Dependence on Experts

Requires skilled security professionals, creating resource bottlenecks

Limited

skilled workforce in cybersecurity



Regulatory & Operational Impact

Non-compliance can lead to penalties, service disruptions and reputational damage

Higher

operational and regulatory risk

Risk & Uncertainty



CIS

**Configuration
Compliance
Automator**

Unified | AI-Assisted
Deterministic | Explainable



To
**Secure & Digital
Infrastructure
Resilient**



BENEFITS

Tangible Advantages for Key Stakeholders



Compliant Systems.
Confident Operations.



For Security Operations Teams

- Faster and accurate compliance checks
- Clear Pass / Fail / Unknown with evidence
- Reduced manual effort and human error



**Hours to
Minutes**

faster compliance audits



For Government / Regulators

- Consistent evaluation across CIS / NIST / STIG
- Transparent and auditable evidence
- Supports national cybersecurity initiatives



**Stronger
Governance**

More resilient national infrastructure



For IT Organizations

- Lower audit preparation time and cost
- Proactive identification and remediation
- Scalable to multi-vendor environments



**Reduced
Operational Cost**

Optimized resources and productivity



For Network Administrators

- Actionable remediation templates (Jinja2)
- Plain-language explanations
- Easier adoption with simple web interface



**Simpler
Operations**

Guided and efficient remediation



For Citizens / Society

- More secure and reliable digital services
- Reduced risk of service disruptions
- Supports a safer and digitally stronger India



Greater Trust

Secure systems.
Stronger Bharat.

FROM
MANUAL EFFORT
AND SECURITY GAPS

AUTOMATE

STANDARDIZE

EXPLAIN

SECURE

BRIDGING THE GAP TO A SAFER TOMORROW

TOWARDS
A SECURE,
DIGITALINDIA

FIELD & APPLIED RESEARCH | From Real-World Insights to Practical Automation

- ✓ Studied **Global Procurement Organization (GPO)** workflows in IT departments and the challenges in standardizing CIS benchmarks.
- ✓ Collected and analyzed **real-world CiS benchmark files** (Windows, domain, security, etc.) to map parameters and create automation rules.
- ✓ Reviewed **CiS official documentation and best practices** for benchmark configuration, validation and policy compliance.



KEY ACADEMIC & GOVERNMENT SOURCES | Trusted References. Stronger Foundations.

01		Center for Internet Security (CIS) Benchmarks <i>Official benchmark documents for Windows, Active Directory and enterprise systems</i>	 https://www.cisecurity.org/cis-benchmarks
02		DISA Security Technical Implementation Guides (STIG) <i>DoD-grade configuration security requirements for network infrastructure devices</i>	 https://public.cyber.mil/stigs/
03		NIST SP 800-53 Security Controls <i>Federal control catalog (e.g. AC-17 Remote Access) mapped to internal compliance rules</i>	 https://csrc.nist.gov/pubs/sp/800/53/r5/upd1/final
04		Cisco IOS/IOS-XE Configuration Fundamentals <i>Scripting and automation for GPO management.</i>	 https://www.cisco.com/c/en/us/support/ios-nx-os-software/ios-15-2m-t/tsd-products-support-series-home.html
05		Multi-Vendor CLI & Config Documentation <i>Vendor configuration references (Junos, FortiOS, PAN-OS) guiding parser normalization to CSM</i>	 https://www.juniper.net/documentation/
06		NIST Cybersecurity Framework <i>Guidelines for secure configuration and compliance.</i>	 https://www.nist.gov/cyberframework
07		NIST SCAP — Security Content Automation Protocol <i>Federal standard for automated compliance checking against CIS/NIST/STIG — validates your rule-engine approach</i>	 https://csrc.nist.gov/projects/security-content-automation-protocol

WHY THESE REFERENCES MATTER

-  **Ensures standardization**
Every configuration is based on official CIS benchmarks.
-  **Provides trusted and proven settings**
Derived from industry and government recognized sources.
-  **Enables reliable automation**
Using well-documented and widely adopted tools.
-  **Builds trust among stakeholders**
Accepted by IT administrators, departments and regulatory bodies.
-  **Keeps the solution up-to-date**
Aligned with the latest benchmark versions and evolving security guidelines.